

INTERNSHIP TASK REPORT
ON
MOTION DETECTION SYSTEM USING
ARDUINO UNO
Task-2

Submitted By

Akuri Pranathi

Intern ID: MT5004



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Maincrafts Technologies

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INTRODUCTION

Embedded systems are small electronic systems that are designed to perform a particular task. These systems contain both hardware and software components working together. Embedded systems are used in many real-life applications such as automatic doors, washing machines, traffic lights, security systems, medical equipment, and smart home devices. The main purpose of embedded systems is to make devices work automatically and efficiently.

Arduino UNO is a popular microcontroller board used for learning electronics and embedded systems. It is easy to use, easy to program, and suitable for beginners. Arduino can be connected with different sensors and output devices to create automation projects. In this project, Arduino UNO is connected with a PIR motion sensor and an LED using the Wokwi simulator.

A PIR (Passive Infrared) sensor is used to detect motion or movement of humans and objects. The sensor works by detecting infrared radiation produced by the human body. When a person moves in front of the sensor, the PIR sensor sends a signal to the Arduino UNO. The Arduino reads this signal and performs an action based on the program written in the code.

In this project, when motion is detected by the PIR sensor, the Arduino turns ON the LED and displays a message on the Serial Monitor. When there is no motion, the LED remains OFF. This project helps in understanding the basic concept of sensor interfacing, motion detection, and automation using embedded systems. Motion detection systems are commonly used in security alarms, automatic lighting systems, smart homes, and surveillance applications.

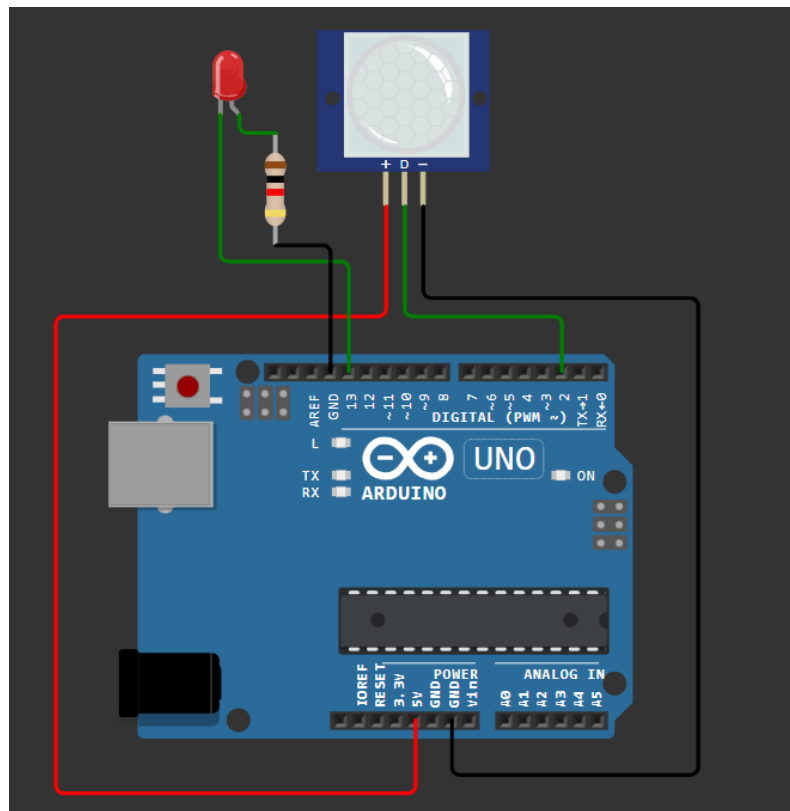
OBJECTIVE

To interface a PIR motion sensor with Arduino UNO and detect motion using LED indication. The project helps in understanding sensor interfacing, Arduino programming, and basic automation using embedded systems.

COMPONENTS REQUIRED

S.NO	COMPONENT
1	Arduino UNO
2	PIR Motion Sensor
3	LED
4	Resistor
5	Jumper Wires
6	Wokwi Simulator

CIRCUIT DIAGRAM



WORKING PRINCIPLE

The working principle of this project is based on motion detection using a PIR (Passive Infrared) sensor and Arduino UNO. The PIR sensor detects infrared radiation emitted by the human body when a person moves near the sensor. When motion is detected, the sensor sends a HIGH signal to the Arduino UNO through the output pin.

The Arduino continuously reads the signal from the PIR sensor. If the signal is HIGH, the Arduino turns ON the LED and displays the message "Motion Detected" on the Serial Monitor. If there is no movement, the PIR sensor sends a LOW signal to the Arduino, and the LED remains OFF.

This project demonstrates how sensors and microcontrollers work together in embedded systems to perform automatic actions. Motion detection systems like this are commonly used in security alarms, automatic doors, smart lighting systems, and home automation applications.

ARDUINO CODE:

```
int pirPin = 2;    // PIR sensor connected to pin 2

int ledPin = 13;   // LED connected to pin 13

int motion;        // Variable to store motion status

void setup() {

    pinMode(pirPin, INPUT);    // Set PIR pin as input

    pinMode(ledPin, OUTPUT);    // Set LED pin as output

    Serial.begin(9600);        // Start serial communication
}

void loop() {

    motion = digitalRead(pirPin); // Read sensor value

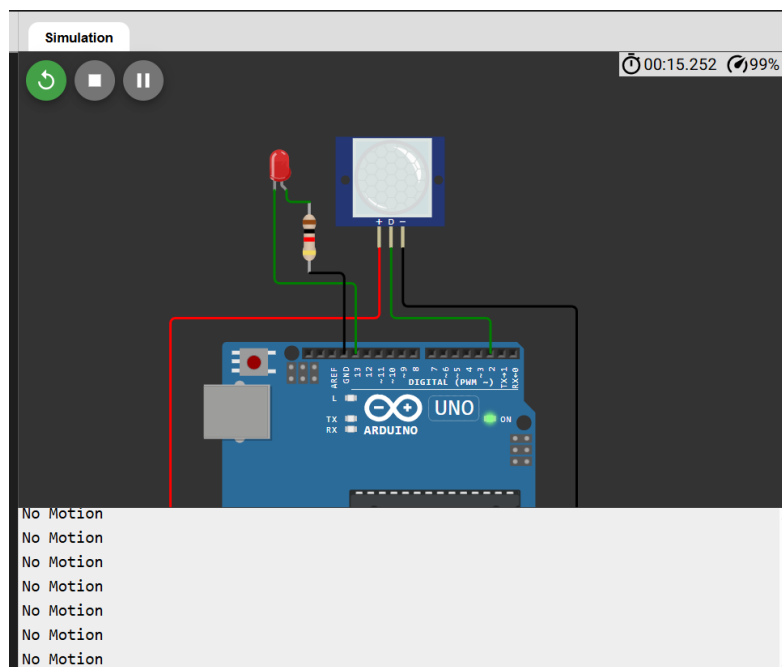
    if (motion == HIGH) {        // Check if motion detected

        digitalWrite(ledPin, HIGH); // Turn ON LED

        Serial.println("Motion Detected!"); // Print message
    }
}
```

```
}else {  
  
    digitalWrite(ledPin, LOW); // Turn OFF LED  
  
    Serial.println("No Motion"); // Print message  
  
}  
  
delay(1000); // Wait for 1 second  
  
}
```

OUTPUT



APPLICATIONS

- Security alarm systems
- Automatic lighting systems
- Smart home automation
- Motion sensing doors
- Surveillance and monitoring systems
- Office and room security systems
- Automatic street lights and smart devices

CONCLUSION

The Motion Detection System using Arduino UNO and PIR sensor was successfully implemented using the Wokwi simulator. The PIR sensor was able to detect motion and send signals to the Arduino. Based on the sensor input, the Arduino controlled the LED and displayed messages on the Serial Monitor. This project helped in understanding sensor interfacing, Arduino programming, and basic automation concepts used in embedded systems and security application